delays, we could extract z values from a standardized normal distribution. In turn, if we would like to challenge Eq. (3), we can collect empirical data of $\vec{x}$ and $\log y$ for different activities, substitute the values in Eq. (4) and compute the z distribution. If our postulate is correct, that distribution should be a standardized normal distribution.

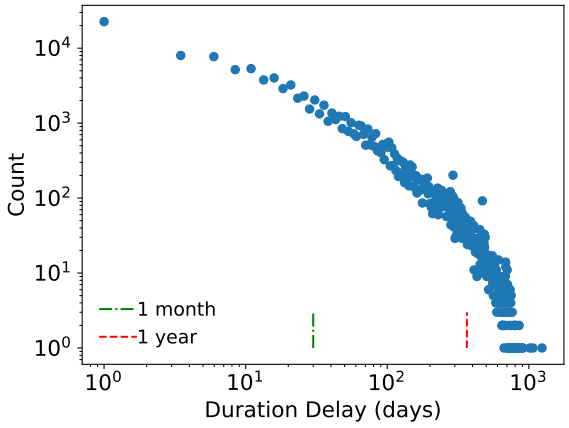


Figure 1: The distribution of delays across 108,745 delayed activities from 180 construction projects.

4. Delay Magnitude

Before digging into possible delay factors, let's have a look at the distribution of delays without filters (Fig. 1). It is evident that delays expand a broad range, from days to years. We could speculate about the shape of this distribution. Yet, there is no point to do so. It represents aggregate data of activities of potentially different types.

5. Delay factors

In this section we uncover delay factors. We gradually increase the number of factors and investigate the convergence of the residual z to the standardized normal distribution. To estimate parameters and validate the model, we pulled together data from 180 construction project schedules (see Appendix Appendix A). The aggregate dataset has about 300,000 actualized activities, including 100,000 delayed activities. This data was used to